node *centrality*

introduction to *network analysis* (*ina*)

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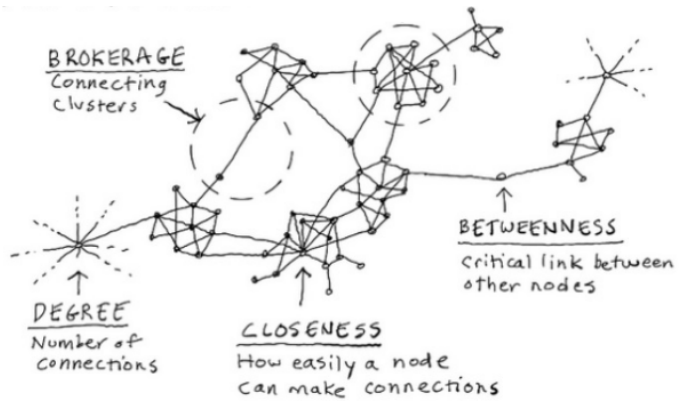
centrality *measures*

which *nodes* are most *important*?

— *node centrality measures* for (*un*)*directed* networks

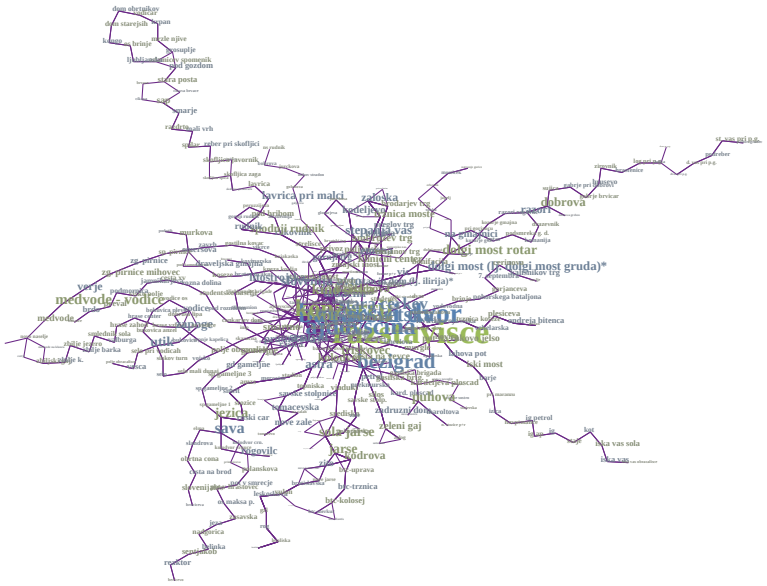
- *clustering coefficients* [WS98, SV05, dNMB05]
- *geodesic-based* measures [Fre77, FBW91, New05]
- *spectral analysis* measures [Kat53, Bon87, BP98]
- *fragment-based* measures [MSOI+02, Prž07, EK15]



— *link analysis algorithms* primarily for *directed* networks

networkology *LPP*

- partial *LPP public bus transport network**
- $n = 416$ bus stops with $\langle k \rangle = 5.62$ connections
- *giant component* 95.4% nodes (6 components)
- *"small-world"* with $\langle C \rangle = 0.09$ and $\langle d \rangle = 14.26$
- *"scale-free"* with $\gamma = 2.62$ for cutoff $k_{min} = 5$



* reduced to largest connected component

centrality *clustering*

important *nodes* are *strongly embedded*

— for *undirected* G *clustering coefficient* C [WS98] of i is

- t_i is number of *linked neighbors* or *triangles* of i

$$C_i = \frac{2t_i}{k_i(k_i-1)} \qquad C_i = 0 \text{ for } k_i \leq 1$$

— ω-*corrected clustering coefficient* C^ω [SV05] of i is

- ω_i is *maximum possible* t_i with *respect to* $\{k\}$

$$C_i^\omega = \frac{t_i}{\omega_i} \qquad C_i^\omega = 0 \text{ for } \omega_i = 0$$

— μ-*corrected clustering coefficient* C^μ [Bat19] of i is

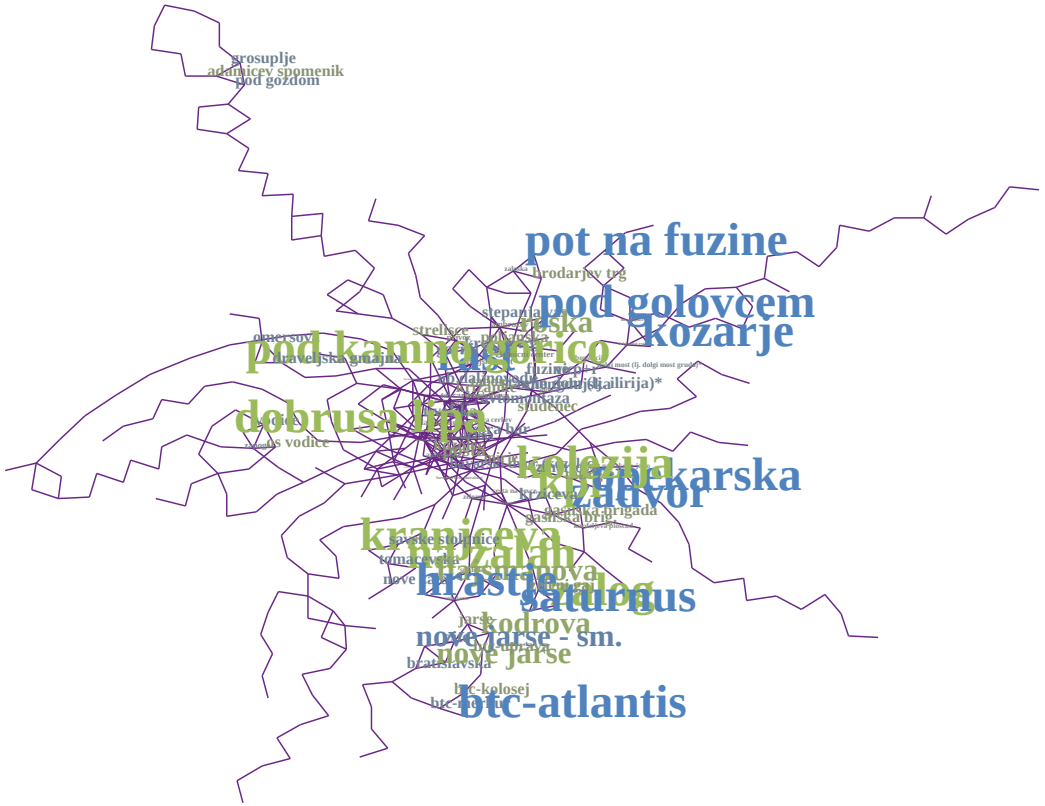
- μ is *maximum* number of *triangles* over *links*

$$C_i^\mu = \frac{2t_i}{k_i\mu} \qquad C_i^\mu = 0 \text{ for } k_i = 0$$

networkology *clustering*

— *clustering coefficient* C in partial LPP network†

— *highest* $C_i = 1.0$ nodes are *Na Žalah etc.* with $k_i = 2$

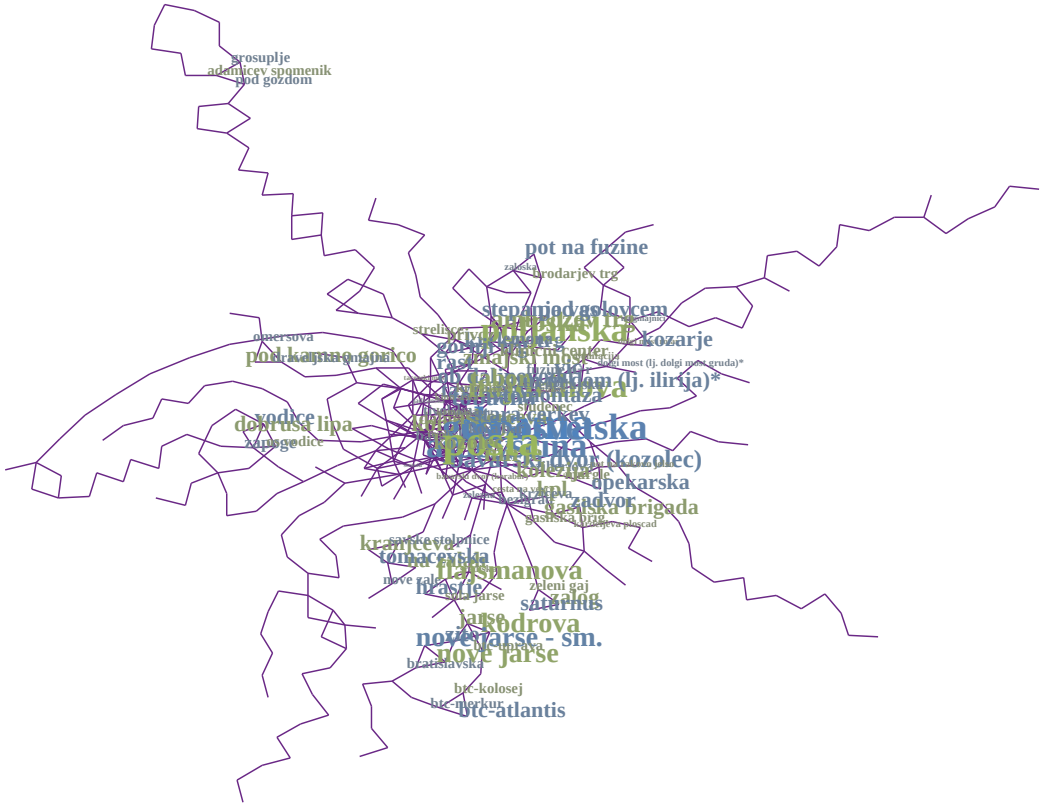


† reduced to simple undirected graph

networkology *μ-clustering*

— *μ-corrected clustering* C^{μ} in partial LPP network[‡]

— *highest* $C_i^{\mu} = 0.44$ node is *Drama* with $k_i = 10$



[‡] reduced to simple undirected graph

centrality *closeness*

important *nodes* are *close to other* nodes

— for (*un*)*directed* G *closeness centrality* ℓ^{-1} [New10] of i is
  - d_{ij} is (*un*)*directed distance* between i and j
  - $d_{ij} = \infty$ for nodes in *different components*

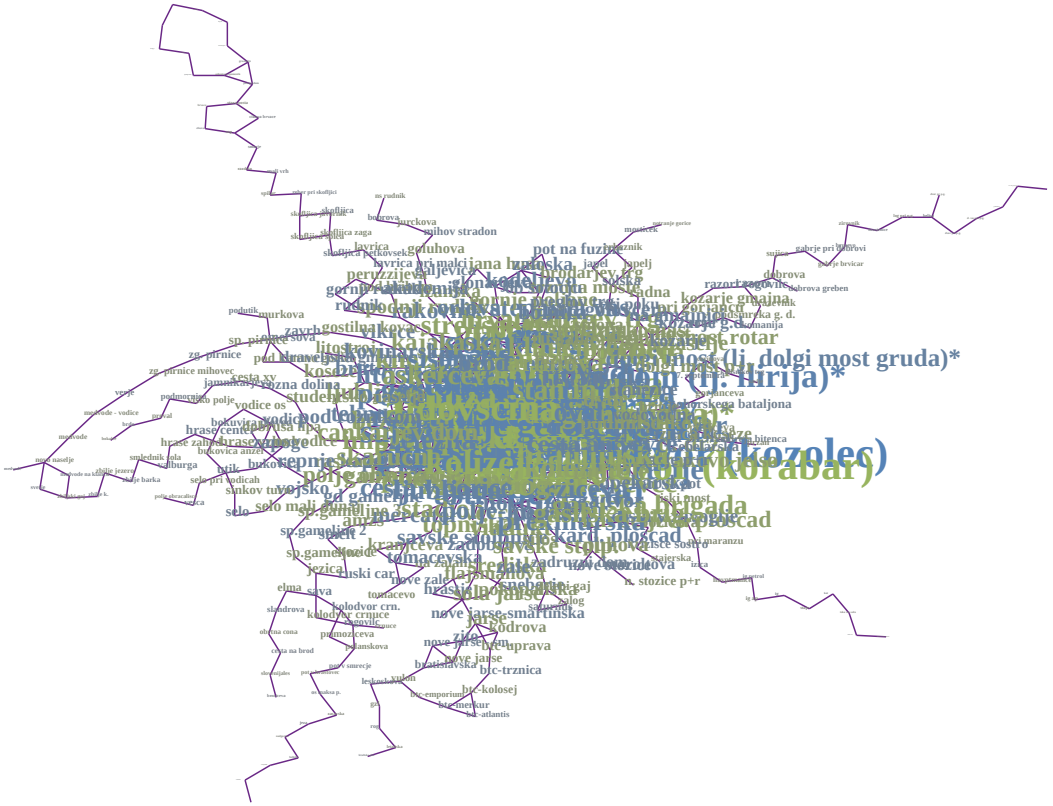
$$\ell_i^{-1} = \frac{1}{n-1} \sum_{j \neq i} \frac{1}{d_{ij}}$$

— ℓ^{-1} spans *small range* in *small-world* networks

networkology *closeness*

— *closeness centrality* ℓ^{-1} in partial LPP network[§]

— *highest* $\ell_i^{-1} = 0.208$ node is *Gosposvetska* with $k_i = 14$



[§] reduced to simple undirected graph

centrality *betweenness*

important *nodes* are *bridges between other* nodes

— for (*un*)*directed* G *betweenness centrality* σ [Fre77] of i is
  - g_{st} is number of *shortest paths between* s and t
  - g_{st}^i is number of *such shortest paths through* i

$$\sigma_i = \frac{1}{n^2} \sum_{st} \frac{g_{st}^i}{g_{st}}$$

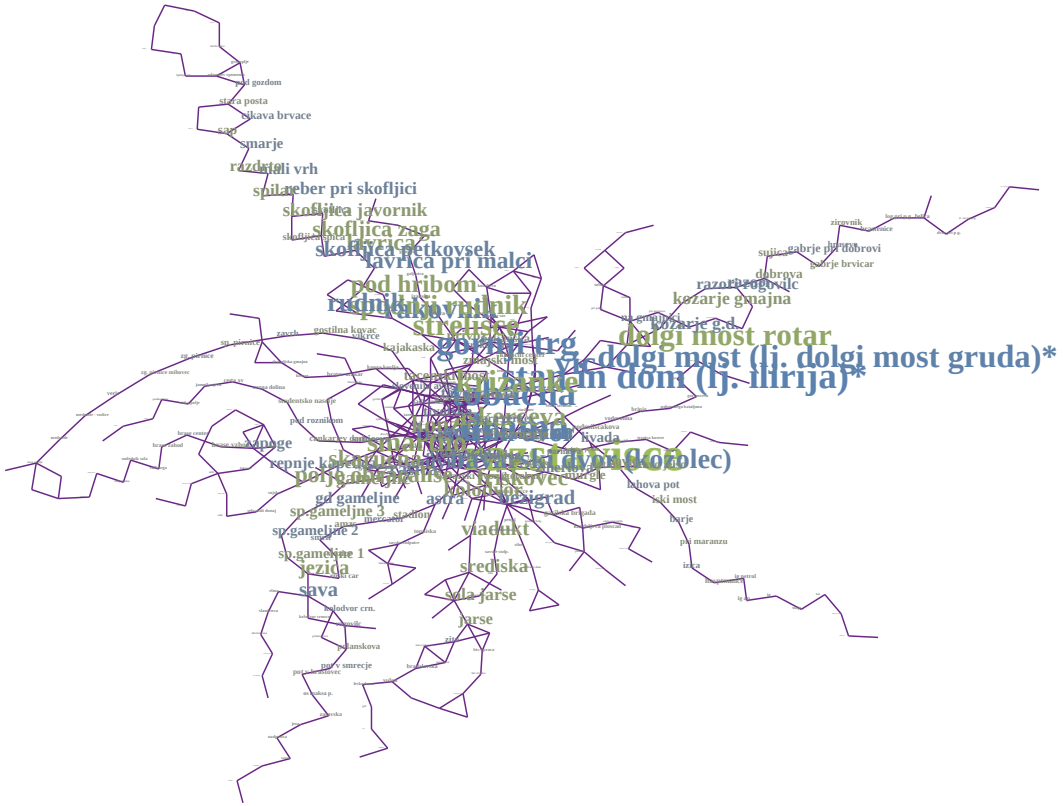
— σ considers *only shortest paths* [FBW91, New05]

— σ mixes *local centers* with *global bridges* [JMK+16]

networkology *betweenness*

— *betweenness centrality* σ in partial LPP network¶

— *highest* $\sigma_i = 0.235$ node is *Razstavišče* with $k_i = 11$


razstavišče
tavrica pri malči
pod hribom
dolgi most rotar
dolgi most (lj. dolgi most gruda)*
vse dom (lj. ilirija)*
kozarje gmajna
razor
zirovnik
gabrje pri dobrovi
gabrje brvicar
dobrova
sujica
skofljica zaga
skofljica netkovšek
skofljica javornik
spilar
reber pri skofljici
smarje
razdrto
sava
jezica
jarse
šola jarše
srediska
viadukt
gd gameljne
sp.gameljne 3
sp.gameljne 2
sp.gameljne 1
asova
vesigrad
kozarje g.d.
izka
pri maranzu
barje
izki most
bohova pot
livada
repnje
snage
pod rozmitom
kajakaska
gostilna kovač
zavrh
sp. gameljne
sp. dobrava
sp. dobrava mihovec
sp. dobrava dolina
sp. dobrava p.
sp. dobrava
sp. dobrava
sp. dobrava

centrality *degrees*

important *nodes* are *linked by many* nodes

— for *undirected* G *degree centrality* d of i is

$$d_i = \frac{1}{n-1} \sum_{j \neq i} A_{ij} = \frac{k_i}{n-1}$$

— in *directed* G *in-degree centrality* d^{in} of i is

$$d_i^{in} = \frac{1}{n-1} \sum_{j \neq i} A_{ij} = \frac{k_i^{in}}{n-1}$$

— in *directed* G *out-degree centrality* d^{out} of i is

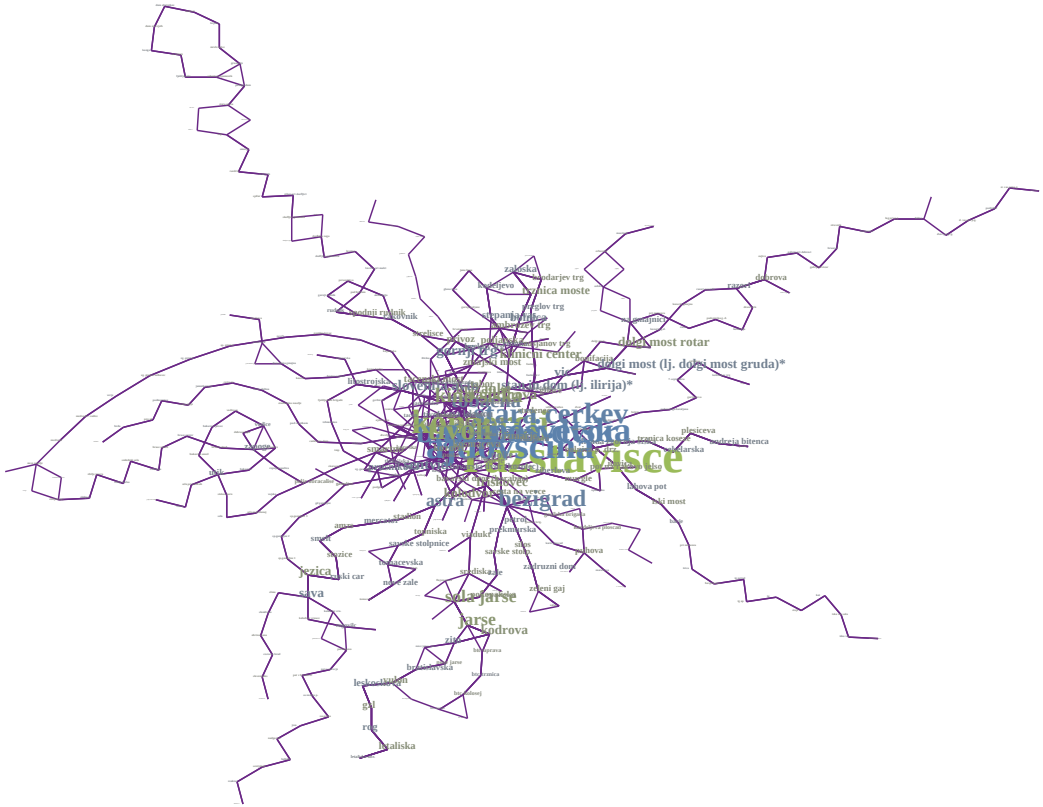
$$d_i^{out} = \frac{1}{n-1} \sum_{j \neq i} A_{ji} = \frac{k_i^{out}}{n-1}$$

networkology *degrees*

— *degree centrality* d in partial LPP network

— *highest* $d_i = 0.099$ node is *Razstavišče* with $k_i = 41$

— *highest* d_i^{in} node is *Razstavišče* with $k_i^{in} = 20$ and $k_i^{out} = 21$



centrality *eigenvector*

important *nodes* are *linked by important* nodes

— for (*un*)*directed* G *eigenvector centrality* e [Bon87] of i is

- v and λ are *eigenvectors* and *eigenvalues* of A
- e is *proportional* to *leading eigenvector* v_1

$$e(t) = A^t e(0) = A^t \sum_i C_i v_i = \sum_i C_i \lambda_i^t v_i = \lambda_1^t \sum_i C_i \left[\frac{\lambda_i}{\lambda_1}\right]^t v_i \to C_1 \lambda_1^t v_1$$

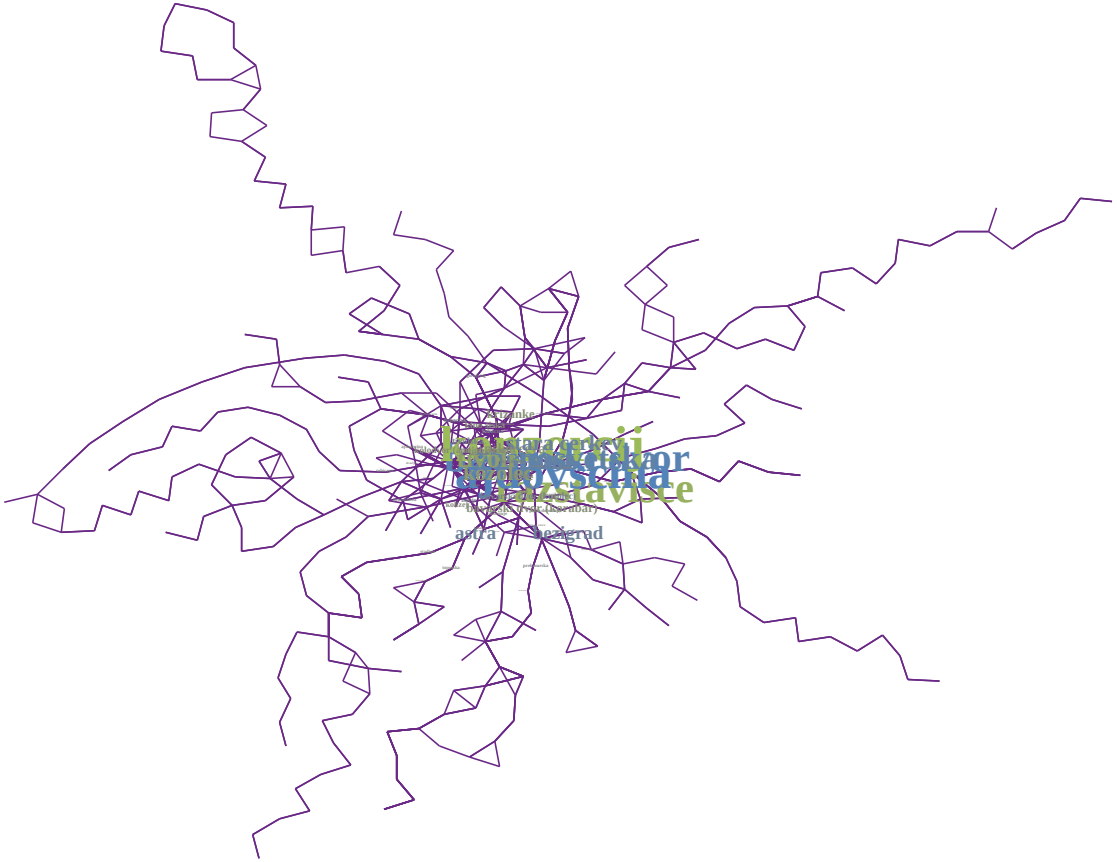
$$e_i = \lambda_1^{-1} \sum_j A_{ij} e_j$$

— in *directed* G $e = 0$ for $k^{in} = 0$ *nodes etc.*

networkology *eigenvector*

— *eigenvector centrality* e in partial LPP network

— *highest* $e_i = 0.082$ node is *Konzorcij* with $k_i = 30$



centrality *Katz*

nodes get small amount of *importance for free*

— for (*un*)*directed* G *Katz centrality* z [Kat53] of i is

- α and β_i are some *positive constants*

$$z_i = \alpha \sum_j A_{ij} z_j + \beta_i$$

— for *convenience* $\beta_i = 1$ whereas $\alpha < \lambda_1^{-1}$

- λ_1 is *leading eigenvalue* of A

centrality *PageRank*

nodes distribute equal amount of *importance*

— for (*un*)*directed* G *PageRank centrality* p [BP98] of i is

- α and β_i are some *positive constants*

$$p_i = \alpha \sum_j A_{ij} \frac{p_j}{k_j^{out}} + \beta_i$$

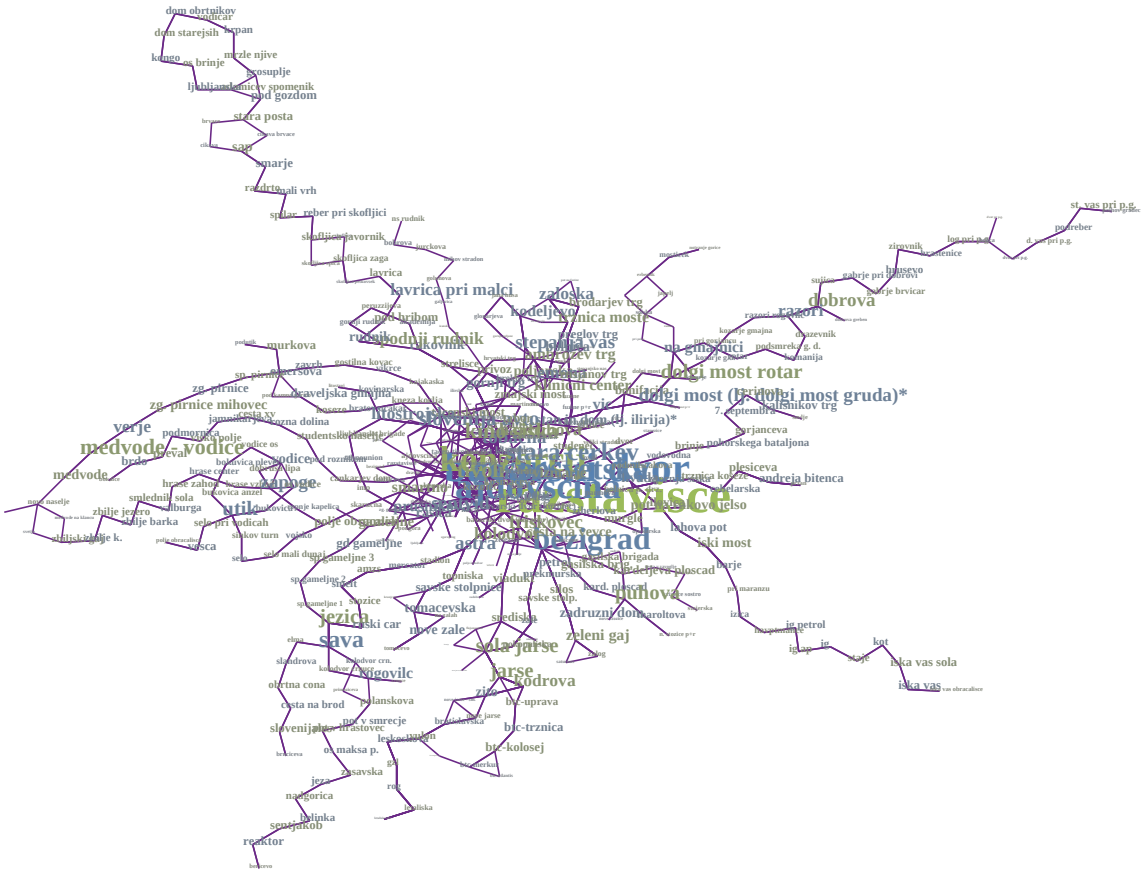
— for *convenience* $\beta_i = \frac{1-\alpha}{n}$ whereas $\alpha = 0.85$

see PageRank algorithm NetLogo demo

networkology *PageRank*

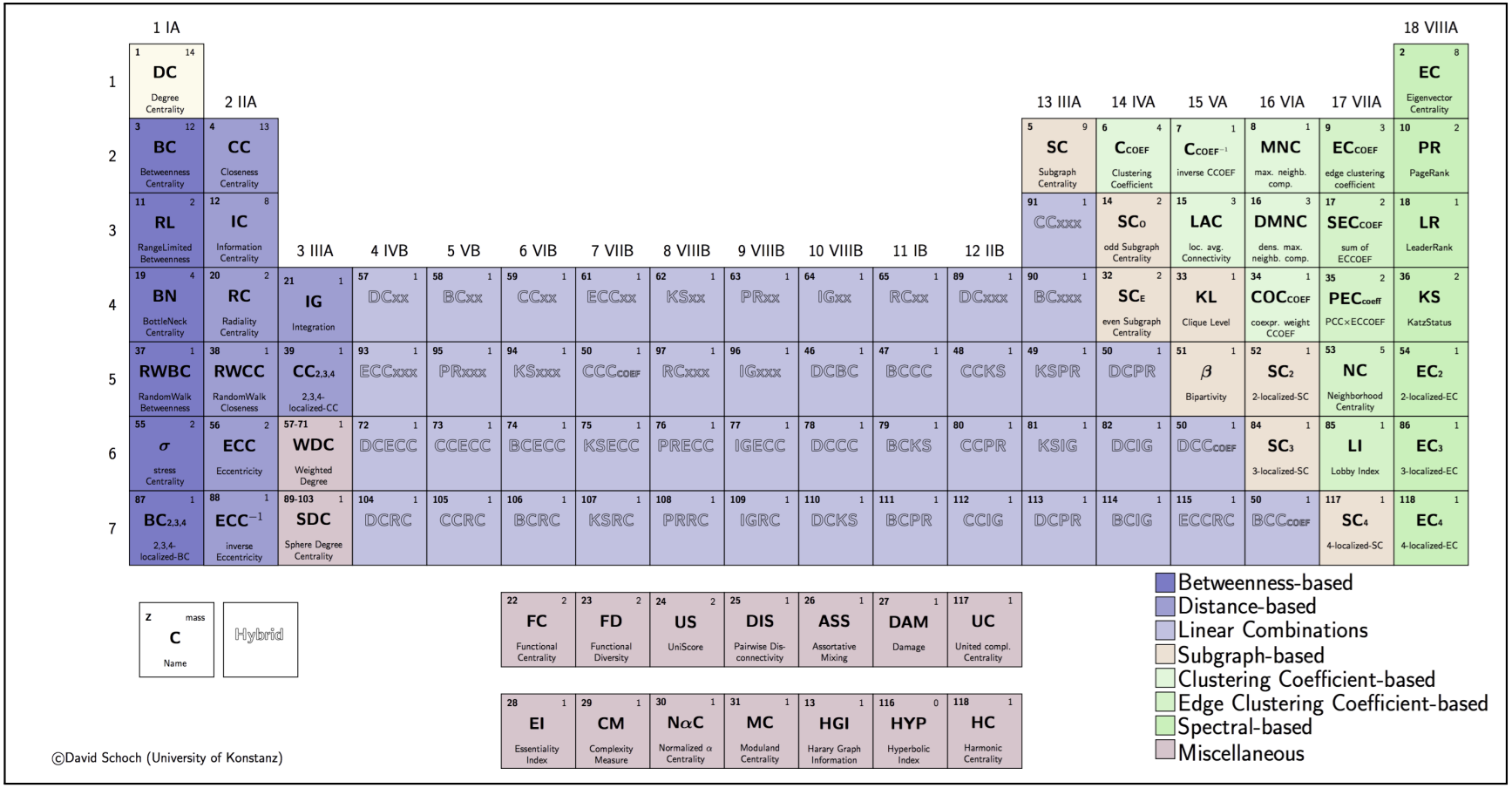
— *PageRank centrality* p in partial LPP network

— *highest* $p_i = 0.011$ node is *Razstavišče* with $k_i = 41$


razstavišče
dolgi most rotar
dobrova
razori
medvode
vodice
verje
sava
jezica
šmarje
lavrica pri malci
zaloška
kodeljevo
stepanja vas
dolgi most (tj. dolgi most gruda)*
bežigrad
šmarna gora
šmarje
šmarjeta
šmarjeta
šmarjeta

centrality *overview*

which *nodes* are most *important*?



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centrality *references*

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